Momentum in two dimensions

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Conserve momentum separately along perpendicular axes.

SUCCESS CRITERIA - I can...

1. Choose and label two axes.
2. Add momentum components along matching directions.
3. Connect component sums to the final motion.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Initial momentum is **0.3 kg·m/s** and measured final momentum is **0.29 kg·m/s**.
1. **Find percent deviation**.
2. **Explain why this** is not itself a measured uncertainty.

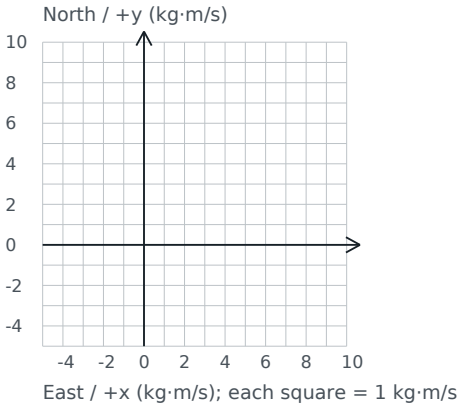
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $p_{x,i} = p_{x,f}$; $p_{y,i} = p_{y,f}$; $p_x = mv_x$; $p_y = mv_y$. Use total system momentum in each direction; negligible external impulse. CPA components are supplied.

Worked reasoning: Two objects have momenta (**3**, **0**) and (**0**, **4**) kg·m/s. Their total is (**3**, **4**), magnitude **5 kg·m/s**. **Add components before finding the** magnitude.

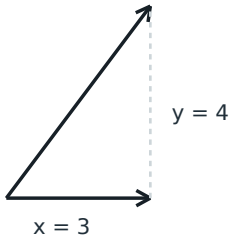
Your turn: Draw a tip-to -tail momentum diagram and **label each axis** with units.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two incoming arrows point *east* and *north*.
1. **Sketch the direction of their** vector sum.
2. Can their ordinary lengths be added for the diagonal magnitude?

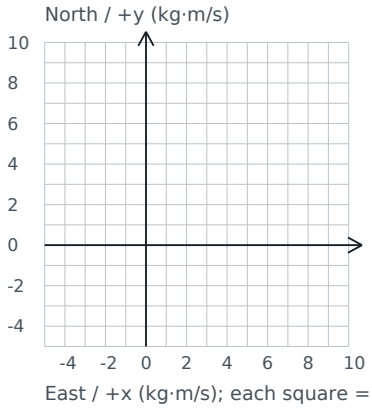


The arrows represent horizontal and vertical components; use their directions to describe the sum.

Marzano check - circle: ① ② ③

04 Apply the physics

A **2 kg** cart at **3 m/s** *east* sticks to a **1 kg** cart at **4 m/s** *north*. **Find final velocity components**, speed and angle *north* of *east*.



Marzano check - circle: ① ② ③

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Continue down each column.

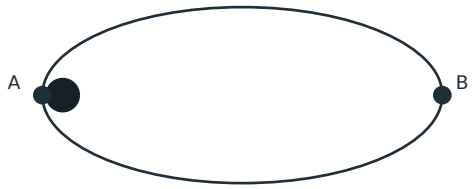
05 Explain your reasoning

Why must both momentum component sums use the same system and same time interval?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

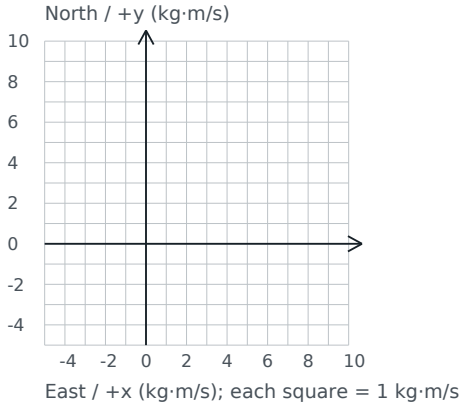
Two ideal Sun-orbits have a_B=**4** a_A. Find T_B/T_A. **Explain why the same normalized** Sun-orbit equation is not automatically valid around another body.
Label the orbital features used in your explanation.



Marzano check - circle: 1 2 3

07 Exit check

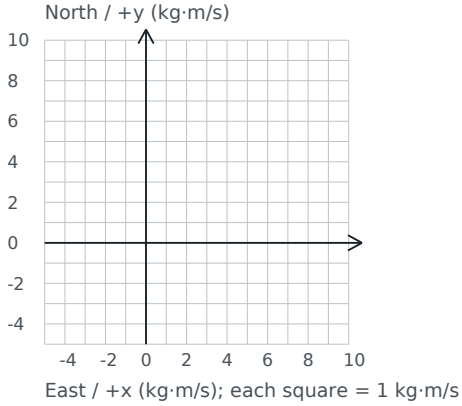
Initial total momentum is (**10**, **0**) kg·m/s. One fragment leaves with (**4**, **3**). **Find the other momentum vector** and its magnitude.



Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Two pucks stick on an ideal air table. One has mass **1 kg** and moves *east* at **4 m/s**. The other has mass **3 kg** and moves *north* at **2 m/s**.
1. **Draw the momentum sum**.
2. **Find the joined pair's velocity** components, speed and direction.



Marzano check - circle: 1 2 3